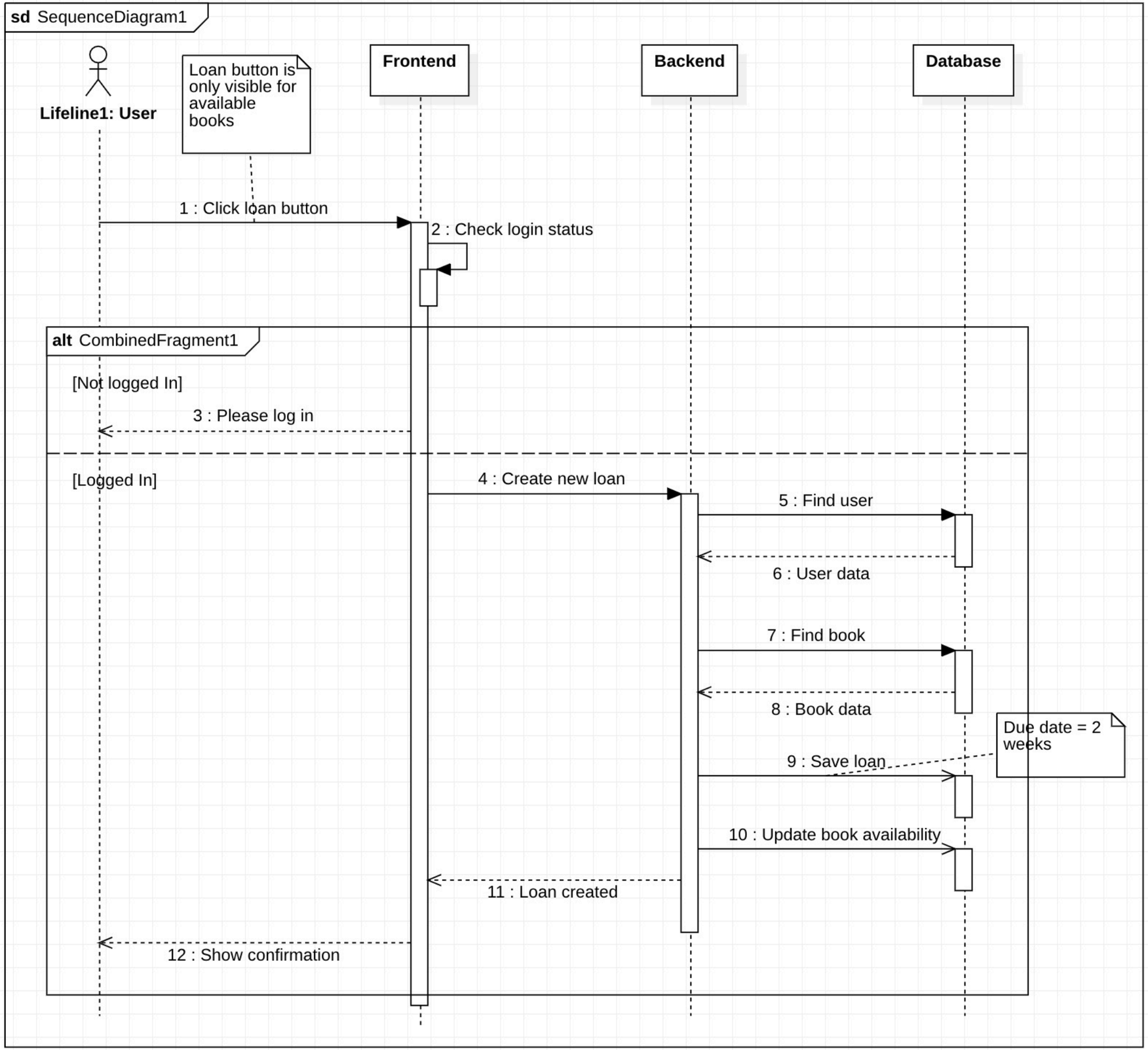

This diagram describes the interaction between the user, frontend, backend and database during the book loan process.

Process begins, when the user clicks the loan button in the frontend. Frontend checks first whether the user is logged in. If not, frontend shows an alert to the user that they need to login to borrow a book. If the user is logged in, frontend sends a request to the backend to create a new loan. Backend fetches user data from the database, then book data and sets the loan due date to two weeks. After this, backend saves the new loan to the database and updates book availability to false. Finally, backend returns a loan confirmation to the frontend and the frontend shows a confirmation alert to the user that the book has been loaned.

Note: the loan button is only visible for available books. so availability is checked before the loan process begins.